System Vision Document

Satellite Ground Station Time Sharing

Problem Description

Egyptian and African universities are building small satellites called CubeSats that need access to ground stations for communication. NSAC currently schedules satellite passes using a spreadsheet, and conflicts happen every week because many universities share a limited network of ground stations. The scheduling process must also consider satellite orbit data, frequency licensing, station capabilities, contact requirements, weather risks, credits, and operational constraints.

It is recommended that a new system be developed and deployed so NSAC can coordinate and share ground-station time fairly and run each scheduled contact. The system should provide a reliable way to calculate available passes, manage contact requests and conflicts, and support communication between the center, remote stations, and university mission operators.

System Capabilities

The new system should be capable of:

- Registering and storing university and satellite information, including orbit data, frequencies, data rate, and mission priority
- Checking frequency licensing and preventing scheduling when a license is invalid
- Downloading fresh TLE orbit data and calculating satellite passes over stations for the next seven days
- Managing ground stations, their supported frequency bands, locations, and status
- Allowing operators to submit contact requests with minimum passes, minimum duration, and mandatory passes
- Detecting conflicts and resolving them by moving a contact, splitting a pass when accepted, or applying mission priority
- Managing contact credits, free passes for new science missions, compensations, and weather-risk flags
- Sending pass plans to remote stations before scheduled contacts and allowing already received plans to execute if the link fails
- Storing telemetry files and linking them to contacts for university access
- Supporting secure remote commanding with target-satellite verification, confirmation, approved access, and command logging

Business Benefits

It is anticipated that the deployment of this new system will provide the following business benefits to NSAC and participating universities:

- Increase fair and transparent use of limited ground-station time among university satellite missions
- Reduce weekly scheduling conflicts and the manual negotiation currently required between operators and universities
- Improve the reliability of satellite contacts and reduce missed communication opportunities
- Provide clearer information about scheduling decisions, including why a contact request was rejected
- Improve operational safety by preventing commands from being sent to the wrong satellite
- Give universities faster access to their telemetry and scheduled pass information
- Provide better monitoring of station usage, successful and failed contacts, credits, and conflict resolution
- Support future growth as additional ground stations and CubeSat missions are added

NATIONAL SPACE & ASTRONAUTICS CONSORTIUM (NSAC)

STAKEHOLDER MAP & REGISTRY

Satellite Ground Station Time Sharing Initiative

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1. Stakeholder Classification Map

Classification	Operational	Executive
Internal	**Eng. Dina Hosny** *Station Operations Lead* **Eng. Ahmed Kamel** *Frequency Coordinator*	**Dr. Yasser Elnaggar** *Ground Segment Manager, Sponsor* **Dr. Samir Ezzat** *Head of Information Security*
External	**Dr. Lamia Rashad** *Mission Operator, Cairo University CubeSat* **Mr. Sani Bello** *Mission Operator, Partner University in Nigeria*	**University / Mission Management** *Participating universities and missions*

2. Comprehensive Stakeholder Registry & System Interests

Category	Stakeholder	Role	Primary Interests & System Requirements
Internal – Operational	**Eng. Dina Hosny**	*Station Operations Lead*	Reliable passes, no missed contacts, station operation, pass plans, and handling scheduling changes.
Internal – Operational	**Eng. Ahmed Kamel**	*Frequency Coordinator*	Ensuring that only satellites with valid licensed frequencies are scheduled.
Internal – Executive	**Dr. Yasser Elnaggar**	*Ground Segment Manager, Sponsor*	Fair use of ground stations, overall scheduling, resource allocation, university usage, and successful operation of the system.

Category	Stakeholder	Role	Primary Interests & System Requirements
Internal – Executive	**Dr. Samir Ezzat**	*Head of Information Security*	Preventing unauthorized satellite control and ensuring secure remote commanding.
External – Operational	**Dr. Lamia Rashad**	*Mission Operator, Cairo University CubeSat*	Getting enough passes, especially after launch, requesting contacts, and accessing telemetry and remote commanding.
External – Operational	**Mr. Sani Bello**	*Mission Operator, Partner University in Nigeria*	Access to ground-station time across time zones and avoiding UTC/time-zone scheduling problems.

Functional Requirements

Satellite Ground Station Time Sharing Initiative

Func Req No	Category	Description / Rules
FR-01	**Registration & Interface**	• The operator shall register each satellite by providing its orbit data, radio frequencies, data rate, and mission priority (science, education, or technology test). • The planning tool shall display all times in the local time zone to avoid confusion with UTC. • The interface shall be readable on a mobile phone.
FR-02	**Frequency Licensing**	• The system shall verify that a satellite's frequencies are licensed. A satellite without a valid license shall not be scheduled. • The license shall be valid on the date of the pass.
FR-03	**Orbit Data Prediction**	• The system shall download TLE orbit data from an external tracking service at least once a day. • Pass prediction for 7 days ahead, covering 15 satellites and 7 stations, shall be completed in under 10 minutes.
FR-04	**Contact Requests**	• Operators shall submit contact requests specifying the minimum number of passes per day, the minimum duration, and any mandatory passes. • A request shall be frozen 24 hours before its pass, except in an emergency. • Emergency mode shall give a satellite in trouble all available passes for 24 hours. The mode shall activate quickly, preferably within one hour.
FR-05	**Conflict Resolution**	• When a conflict occurs, the system shall first try to move one satellite to another station. If this fails, it shall split the pass into two halves, provided both operators accept. If this also fails, the conflict shall be decided by priority. • A satellite that loses a contact shall receive a higher priority in the next conflict, so that no satellite is starved. • When a request is rejected, the system shall show which satellite was given the slot.

Func Req No	Category	Description / Rules
FR-06	**Credits & Budgets**	• Each university shall receive 600 credits per year. One minute of contact shall cost one credit. • A new science mission shall receive 50 free passes during the first 30 days after launch. • Passes lost because of a station failure shall be compensated within the following 7 days. • Passes shall be flagged as a weather risk when dust storms are forecast.
FR-07	**Station Operations**	• Each station shall receive its pass plan at least 30 minutes before the pass. • If the link to the center fails, the station shall still execute the plans it has already received. • Operators shall be notified by email or SMS 15 minutes before each of their passes. • Universities shall be able to export their pass plan to their own calendar.
FR-08	**Telemetry Storage**	• Telemetry shall belong to the satellite owner and shall be kept by the center for 1 year. • Universities shall be able to download their telemetry data at any time, without waiting for an email. • The received telemetry files are expected to be between 5 MB and 200 MB per pass.
FR-09	**Commanding Security**	• Every command shall be signed with the university's key, sent only from approved IP addresses, and protected by two-factor login. • Commanding shall be allowed only during that university's own passes. • Every command shall be logged. • The university's operators shall be able to block commanding immediately.
FR-10	**Command Safety**	• Every command shall contain the ID of the target satellite, which shall be checked against the satellite in view. • A confirmation step shall be required before a command is sent.
FR-11	**System Questions / Reporting**	• How many successful contacts did each satellite get this week? • Which stations are most overloaded? • How many conflicts were solved by moving, splitting or priority? • How many credits has each university used?
FR-12	**Emergency Mode**	• An emergency mode that gives a satellite in trouble all available passes for 24 hours.
FR-13	**Data Sharing**	• A data sharing area where universities publish Earth images for other researchers.

Func Req No	Category	Description / Rules
NFR-01	**Constraints**	• Two new CubeSats launch in eight months. The system must be ready before their first pass. • Commands must be secure, so no university can ever send commands to another university's satellite.

Requirements Elicitation & Interview Agendas

Satellite Ground Station Time Sharing

4. Elicitation Techniques

The project team will use a combination of document analysis, stakeholder interviews, workflow walkthroughs, questionnaire design, and validation discussions. Each technique addresses a different type of information needed to define a consistent scheduling and ground-station time-sharing system.

Technique	Stakeholder / source	Information to obtain	Application to this system	Responsible
Document analysis	Gathered Data. <Recourses>	Business rules, constraints, actors, data, operational problems, security requirements and stated user needs.	Establishes the baseline for scheduling, licensing, station planning, telemetry, credits, failures, weather risk and remote commanding.	Qais
Executive interview	Dr. Yasser Elnaggar — Ground Segment Manager / sponsor	Business objectives, priorities, KPIs, scope, emergency approval, reporting and first-version decisions.	Clarifies how fair allocation and management priorities should influence system behavior.	Mark
Operational interview	Eng. Dina Hosny — Station Operations Lead	Current scheduling workflow, station constraints, late changes, pass-plan delivery, failures and prediction workload.	Defines the operational process the system must support and the exceptions it must handle.	Abdullah
Operational user interviews	Dr. Lamia Rashad; Mr. Sani Bello; plus relevant licensing/security stakeholders	Mission request needs, time-zone requirements, telemetry access, notifications, remote commanding, licensing and security constraints.	Connects the system to the real needs of university and international users and to authorization requirements.	Hazem
Workflow walkthrough	Scheduling and station-operation workflow	Sequence of request submission, validation, conflict resolution, pass assignment, plan delivery, contact execution and result handling.	Helps identify missing steps and exception paths such as station failure, link failure and weather risk.	Abdullah
Questionnaire	University mission operators, student operators and station operators	Frequency of conflicts, manual negotiation, time display, reminders, phone use, calendar export, telemetry, shorter contacts and remote commanding.	Provides structured feedback on usability and operational priorities.	Elsayed
Validation discussion	Responsible NSAC and university stakeholders	Unresolved rules and decisions identified during analysis.	Prevents unsupported assumptions from becoming requirements.	Qais

4.1 Elicitation Focus

Focus area	What is already established	What still needs confirmation
Scheduling and allocation	The system calculates passes, handles conflicts by moving, splitting, then priority, and gives a losing satellite higher priority next time.	Exact priority weights and tie-breaking rules.
Operational timing	Orbit data is refreshed at least daily; prediction covers seven days; prediction must finish under ten minutes; stations receive plans at least 30 minutes before a pass.	Detailed operational workflow and handling of exceptional timing conditions.
User interaction	Users need time-zone display, rejection explanations, notifications, phone-friendly access and calendar export.	Exact notification channels/defaults and detailed UI behavior.

Focus area	What is already established	What still needs confirmation
Telemetry	Telemetry belongs to the satellite owner and is retained by the center for one year; users want self-service download.	Detailed permissions, download limits and audit behavior.
Remote commanding	If enabled, commands require university signing, approved IP, two-factor authentication, own-pass restriction, logging, target ID and confirmation; operators can block it.	First-release scope and complete permission matrix.

5.1 Interview Agendas

The agendas below are designed to validate and clarify information already contained in the project resource pack. They are organized around stakeholder responsibilities, operational activities, information inputs/outputs, exceptions, and decisions that affect system requirements.

5.1 Internal Executive Interview — Dr. Yasser Elnaggar

No.	Question	Type	Source-based context / expected clarification
1	What is the primary problem the system must solve, and what outcome would demonstrate that it has been solved?	Open-ended	The request identifies spreadsheet-based scheduling conflicts and the goal of fair station-time sharing and contact execution.
2	Which measures should management use to evaluate the system?	Open-ended	The project identifies successful contacts, station load, conflict-resolution methods and credits used; a monthly university-usage/failure report is also requested.
3	How should the system prioritize competing contact requests?	Open-ended	The documented sequence is move → split if both accept → priority, with increased priority for a satellite that loses a contact. Exact weights/tie-breaking remain unspecified.
4	Who has authority to approve an emergency request, and how quickly should it be processed?	Closed-ended + follow-up	The mission operator requests; Yasser or Dina approves; the response may need to occur within about one hour.
5	Should requests normally be frozen 24 hours before the pass, with emergencies as the exception?	Closed-ended	Dina requested a 24-hour freeze except emergencies.
6	Which capabilities should be included in the first remote-commanding release?	Open-ended	Remote commanding is intended to be phased; the first-version scope is not defined in the source.
7	What management information should be included in the monthly report?	Open-ended	The source requests university pass usage and failures attributable to NSAC.
8	How should station failures, replacement passes and weather-risk decisions be represented to management and users?	Open-ended	Station-caused failures receive compensation within seven days; dust-storm risk is flagged and the operator decides whether to keep the pass.
9	What institutional priorities should influence the allocation of limited station time?	Open-ended	The project establishes mission priority categories and fair sharing, but does not specify numerical priority weights.
10	What is the required readiness point for the system?	Closed-ended	The system must be ready before the first pass of two CubeSats launching in eight months.

5.2 Internal Operational Interview — Eng. Dina Hosny

No.	Question	Type	Source-based context / expected clarification
1	How are ground-station passes and contact requests handled today, and where does the current process create operational difficulty?	Open-ended	The current process uses a spreadsheet and conflicts occur weekly; manual negotiation can take an hour.
2	What information must be available before an operator can schedule a contact?	Open-ended	The system needs satellite/orbit data, frequencies, data rate, mission priority, station capability and request constraints.
3	Should normal contact requests be frozen 24 hours before a pass? What exceptions are required?	Closed-ended + follow-up	Dina requested a 24-hour freeze except emergencies.
4	What must be included in the pass plan sent to a station?	Open-ended	The plan includes antenna pointing angles and commands to send, and must reach the station at least 30 minutes before the pass.
5	What should the station do if its link to the center fails after receiving a plan?	Open-ended	Remote stations must still execute plans already received.
6	How should the system handle a station-caused contact failure?	Open-ended	The university receives a free replacement pass; policy specifies compensation within the next seven days.
7	How should weather risks be presented and who decides whether to keep an at-risk pass?	Open-ended	Dust-storm forecasts mark passes at risk; the operator decides whether to keep them.
8	What prediction workload and response time must the scheduling process support?	Closed-ended	Seven-day prediction for 15 satellites over 7 stations must finish in under ten minutes; TLE data is downloaded at least daily.
9	When two requests conflict, what information should the operator see before deciding or applying the scheduling rule?	Open-ended	Users want to understand why requests are rejected and which satellite received the slot.
10	What operational status should be recorded after a contact?	Open-ended	Contacts must record assignment, whether moved/split, the conflict solved, and success/failure; telemetry and commands are linked to contacts.

5.3 External Operational User Validation — Lamia Rashad and Sani Bello

No.	Question	Type	Source-based context / expected clarification
1	What information should a university provide when registering a satellite and submitting a contact request?	Open-ended	Registration includes orbit data, radio frequencies, data rate and mission priority; requests include minimum passes/day, minimum duration and mandatory passes.
2	How should the system support the first days after launch?	Open-ended	Lamia needs every possible pass in the first week to verify the satellite is alive; new science missions also receive 50 free passes in their first 30 days. (02 Users Meeting Transcript;
3	Which time representation should be displayed for international users?	Closed-ended	Sani specifically requires local time because UTC confusion caused a missed pass.
4	What notifications should be sent before a pass?	Closed-ended + follow-up	Mission-operator feedback requests an email or SMS 15 minutes before each pass.
5	What information should be shown when a contact request is rejected?	Open-ended	Users want the reason and identification of the satellite that took the slot.
6	What telemetry access is required after a contact?	Open-ended	Telemetry belongs to the satellite owner, is kept by the center for one year, and Lamia requests self-service download.
7	Would calendar export improve your workflow?	Closed-ended	Mission-operator feedback explicitly requests export of the pass plan to the user's calendar.

No.	Question	Type	Source-based context / expected clarification
8	If a full contact cannot be allocated, under what conditions would a shorter contact be acceptable?	Open-ended	The system may split a conflicting pass if both parties accept shorter contacts.

5.4 Supporting Operational Validation — Ahmed Kamel and Dr. Samir Ezzat

No.	Question	Type	Source-based context / expected clarification
1	When is a satellite eligible to be scheduled with respect to frequency licensing?	Closed-ended	Scheduling is permitted only when frequencies are licensed and the license is valid on the pass date.
2	How should an invalid or pending license status affect the scheduling workflow?	Open-ended	The supplied files state the eligibility rule but do not specify the detailed user notification/status behavior.
3	If remote commanding is enabled, what controls are mandatory?	Open-ended	University-signed commands, approved IP addresses, own-pass restriction, two-factor authentication, command logging and immediate operator blocking are required.
4	How must the system prevent commands from reaching the wrong satellite?	Open-ended	Commands require target satellite ID, verification against the satellite in view and a confirmation step.
5	What remote-commanding capabilities belong in the first version?	Open-ended	The decision is to enable remote commanding in phases; exact first-release capabilities remain to be defined.

5.5 Operational User Questionnaire

Purpose: capture structured experiences and preferences from university mission operators, student operators and station operators. The questionnaire is a proposed instrument; no responses are treated as collected results.

No.	Question	Response options	Requirement area
1	Which role best describes your work?	University mission operator / Student operator / Station operator / Other	User profile
2	During the past 30 days, how many contact requests or scheduled passes have you worked with?	0 / 1–5 / 6–10 / 11–20 / More than 20 / Not sure	Usage
3	How often did a scheduling conflict disrupt your work?	Never / Once / 2–3 times / 4 or more / Not sure / N/A	Conflict management
4	How often did you need manual negotiation or follow-up to resolve a conflict?	Never / Once / 2–3 times / 4 or more / Not sure / N/A	Conflict workflow
5	How clear are explanations when a request is rejected or changed?	Very unclear → Very clear / Have not received one / N/A	Transparency
6	Which device do you use most often for pass planning?	Desktop/laptop / Phone / Tablet / Other / N/A	Mobile usability
7	Which time display best supports your work?	Local / UTC / Both / No preference / N/A	Time-zone handling
8	What is your preferred reminder before a pass?	Email / SMS / Both / No reminder / Other / N/A	Notifications
9	Rate the importance of phone-friendly planning, calendar export, telemetry download and clear rejection explanations.	1 Not important → 5 Essential / N/A	Usability & access
10	If a full contact cannot be allocated, when could a shorter contact meet your needs?	Usually / Sometimes / Rarely / Never / Not sure / N/A	Conflict resolution
11	How important is direct remote commanding from your university lab?	Not important → Essential / N/A	Remote commanding
12	What is the single most important improvement or missing need?	Open response	Additional requirements

NATIONAL SPACE & ASTRONAUTICS CONSORTIUM (NSAC)

WORKFLOW DOCUMENTATION

Satellite Ground Station Time Sharing

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1. Introduction

This document records two complex business processes of the Satellite Ground Station Time Sharing system. Each process is modelled as a UML activity diagram with swim lanes, so that it is clear who performs every action, where decisions are taken and how the process ends. A short written description follows each diagram.

The first process covers how a mission operator's contact request becomes a scheduled contact, including licence checking and conflict resolution. The second covers how the control center prepares a station plan and how the ground station executes it, including the case where the link to the control center is lost.

2. Process Overview

Business Process	Purpose	Participants (Swim Lanes)
Contact Scheduling & Conflict Resolution	Turns a contact request into a scheduled contact, checking the frequency licence and station availability and resolving any conflict by moving, splitting or applying priority.	Mission Operator; NSAC System; Frequency Coordinator (Ahmed Kamel); Affected Mission Operator
Station Plan Preparation & Contact Execution	Prepares the station plan for a scheduled contact, sends it ahead of the pass and makes sure the station can execute it even if the link to the control center is lost.	NSAC Operations / Control Center; Ground station (second swim lane)

3. Activity Diagram 1: Contact Scheduling & Conflict Resolution

The diagram shows four swim lanes: the Mission Operator, the NSAC System, the Frequency Coordinator (Ahmed Kamel) and the Affected Mission Operator.

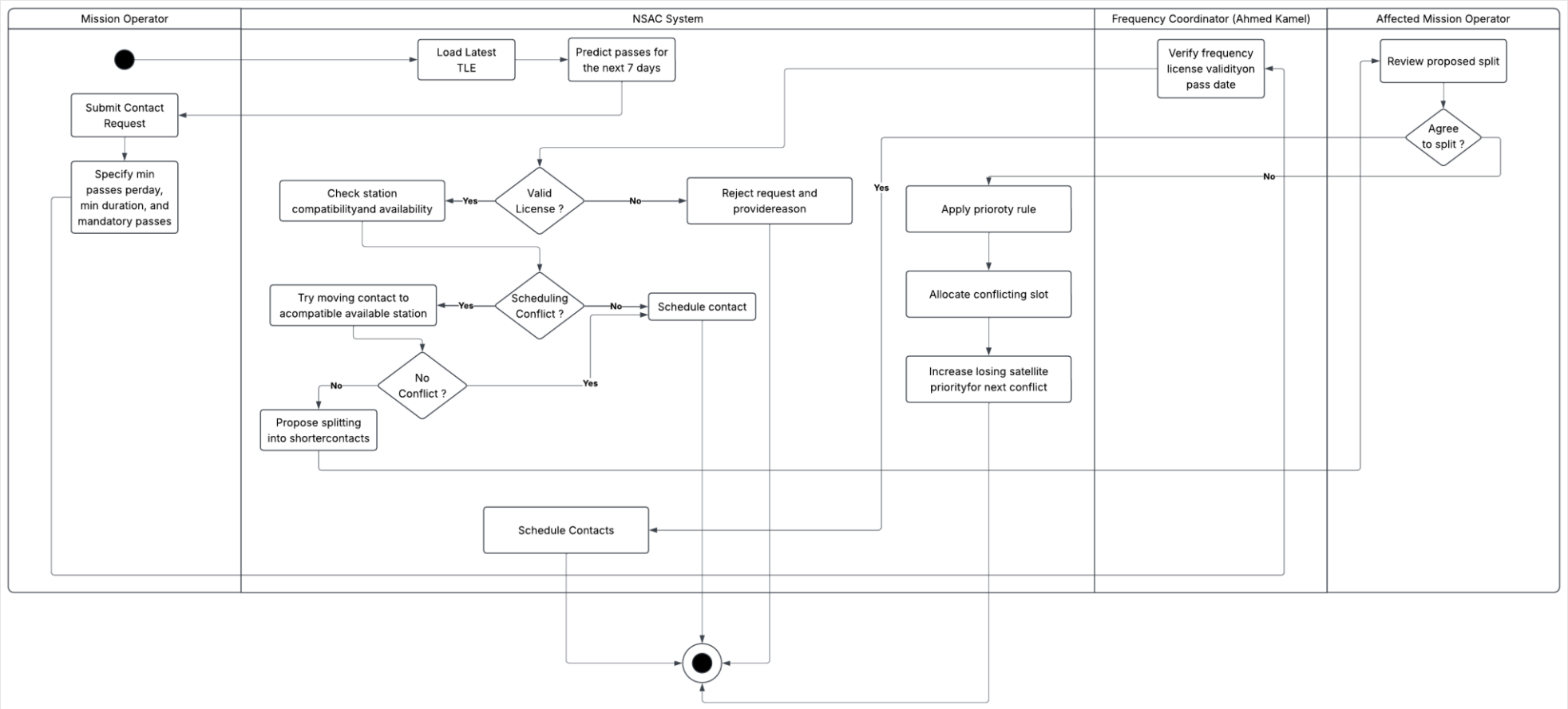


Figure 1: Activity Diagram 1 – Contact Scheduling & Conflict Resolution

3.1 Process Description

1. The process starts when the NSAC System loads the latest TLE data and predicts passes for the next 7 days.
2. The Mission Operator submits a contact request and specifies the minimum passes per day, the minimum duration and the mandatory passes.
3. The Frequency Coordinator verifies the validity of the frequency license on the pass date.
4. If the license is not valid, the system rejects the request and provides the reason, and the process ends.
5. If the license is valid, the system checks station compatibility and availability, then checks for a scheduling conflict.
6. If there is no conflict, the system schedules the contact and the process ends.
7. If there is a conflict, the system tries moving the contact to a compatible available station. If the conflict is resolved, the contact is scheduled.
8. If the conflict remains, the system proposes splitting the request into shorter contacts, and the Affected Mission Operator reviews the proposed split.
9. If the operator agrees to the split, the system schedules the contacts and the process ends.
10. If the operator does not agree, the system applies the priority rule, allocates the conflicting slot and increases the priority of the losing satellite for the next conflict. The process then ends.

4. Activity Diagram 2: Station Plan Preparation & Contact Execution

The diagram shows two swim lanes: NSAC Operations / Control Center, which prepares and sends the plan, and the ground station lane, which receives and executes it.

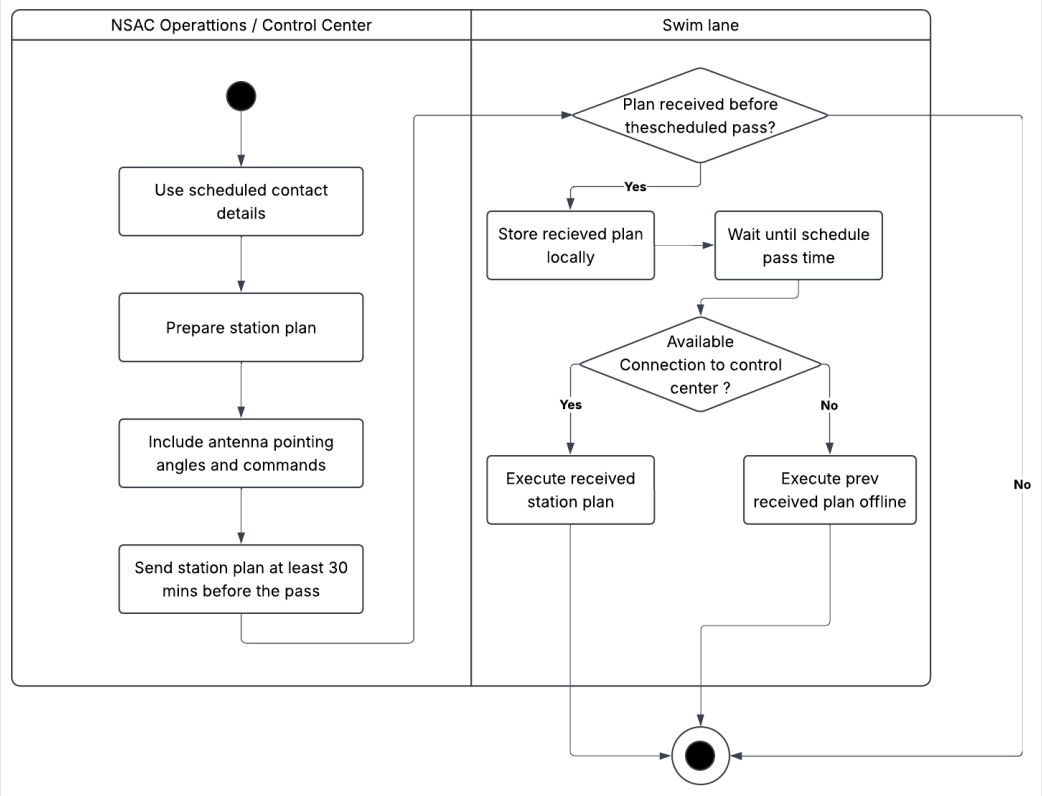


Figure 2: Activity Diagram 2 – Station Plan Preparation & Contact Execution

4.1 Process Description

1. The control center uses the scheduled contact details and prepares the station plan.
2. The plan includes the antenna pointing angles and the commands to be executed.
3. The control center sends the station plan at least 30 minutes before the pass.
4. The station checks whether the plan was received before the scheduled pass. If it was not, the process ends.
5. If the plan was received, the station stores it locally and waits until the scheduled pass time.
6. At pass time, if the connection to the control center is available, the station executes the received station plan.
7. If the connection is not available, the station executes the previously received plan offline, so the contact is not lost. The process then ends.